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By Amit Katiyar (MCA-JNU)



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TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)

Inverse Trigonometry

NIMCET PYQ



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1. If $\theta = \tan^{-1} \left[\frac{1}{1+2} \right] + \tan^{-1} \left[\frac{1}{1+(2)(3)} \right] + \tan^{-1} \left[\frac{1}{1+(3)(4)} \right] + \dots + \tan^{-1} \left[\frac{1}{1+n(n+1)} \right]$, then $\tan \theta$ is equal to

(a) $\frac{n}{n+1}$ (b) $\frac{n+1}{n+2}$ (c) $\frac{n}{n+2}$ (d) $\frac{n-1}{n+2}$

NIMCET 2009

2. The number of solutions for $\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$ is/are

(a) -2 (b) 1 (c) -1 (d) 2

NIMCET 2009

3. If $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$ then x is

(a) 1/6 (b) 1/3 (c) 1/2 (d) 1/4

NIMCET 2009

4. If $\sin^{-1} x + \cos^{-1}(1-x) = \sin^{-1}(-x)$, then x satisfies the equation

(a) $2x^2 - x + 2 = 0$ (b) $2x^2 - 3x = 0$
(c) $2x^2 + x - 1 = 0$ (d) None

NIMCET 2009

5. If $\sin^{-1} \left(\frac{2a}{1+a^2} \right) - \cos^{-1} \left(\frac{1-b^2}{1+b^2} \right) = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ then is equal to

(a) a (b) b (c) $\frac{a+b}{1-ab}$ (d) $\frac{a-b}{1+ab}$

NIMCET 2010

6. The value of $\cot^{-1}(21) + \cot^{-1}(13) + \cot^{-1}(-8)$ is

(a) 0 (b) π (c) ∞ (d) $\frac{\pi}{2}$

NIMCET 2012

7. The value of $\sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2}-\sqrt{1}}{\sqrt{6}} + \sin^{-1} \frac{\sqrt{3}-\sqrt{2}}{\sqrt{12}} + \dots$ to infinity is equal to

(a) π (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

NIMCET 2015

8. The principal value of $\cot^{-1}(-\sqrt{3})$ is

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{6}$ (c) $\frac{7\pi}{6}$ (d) $\frac{5\pi}{6}$

NIMCET 2017

9. The value of $\cot \left(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3} \right)$ is

(a) $\frac{6}{17}$ (b) $\frac{3}{17}$ (c) $\frac{4}{17}$ (d) $\frac{5}{17}$

NIMCET 2018

10. The value of $2 \tan^{-1} [\operatorname{cosec}(\tan^{-1} x) - \tan(\cot^{-1} x)]$ is

(a) $\tan x$ (b) $\cot x$
(c) $\tan^{-1} x$ (d) $\operatorname{cosec}^{-1} x$

NIMCET 2020

11. The value of $\cot \left(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3} \right)$ is

(a) $\frac{6}{17}$ (b) $\frac{3}{17}$ (c) $\frac{4}{17}$ (d) $\frac{5}{17}$

NIMCET 2022

12. If $\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} = \phi$, then $9x^2 - 12xy \cos \phi + 4y^2$ is equal to

(a) $-36 \sin^2 \phi$ (b) $36 \sin^2 \phi$
(c) $36 \cos^2 \phi$ (d) 36

NIMCET 2022

13. The correct expression for $\cos^{-1}(-x)$ is

(a) $\frac{\pi}{2} - \cos^{-1} x$ (b) $\pi - \cos^{-1} x$
(c) $\pi + \cos^{-1} x$ (d) $\frac{\pi}{2} + \cos^{-1} x$

NIMCET 2022

14. Solutions of the equation $\tan^{-1} \sqrt{x^2 + x} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$ are

(a) 0, 1 (b) 1, -1 (c) 0, -1 (d) 0, -2

NIMCET 2022

15. If n_1 and n_2 are the number of real valued solutions $x = |\sin^{-1} x|$ and $x = \sin(x)$ respectively, then the value of $n_2 - n_1$ is

(a) 1 (b) 0 (c) 2 (d) 3

NIMCET 2023

ANSWER KEY

1.	c	2.	c	3.	a	4.	d	5.	d
6.	b	7.	c	8.	d	9.	a	10.	c
11.	a	12.	b	13.	b	14.	c	15.	b

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	00
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	04	NIMCET 2018	01
NIMCET 2010	01	NIMCET 2019	00
NIMCET 2011	00	NIMCET 2020	01
NIMCET 2012	01	NIMCET 2021	00
NIMCET 2013	00	NIMCET 2022	04
NIMCET 2014	00	NIMCET 2023	01
NIMCET 2015	01	NIMCET 2024	00
		NIMCET 2025	00

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SOLUTION

1. (c) $\theta = \tan^{-1}\left(\frac{1}{1+2}\right) + \tan^{-1}\left(\frac{1}{1+(2)(3)}\right)$
 $+ \tan^{-1}\left(\frac{1}{1+(3)(4)}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right)$

$$\theta = \tan^{-1}\left(\frac{2-1}{1+2}\right) + \tan^{-1}\left(\frac{3-2}{1+(2)(3)}\right)$$

$$+ \tan^{-1}\left(\frac{4-3}{1+(3)(4)}\right) + \dots + \left(\frac{n+1-n}{1+n(n+1)}\right)$$

$$\theta = (\tan^{-1} 2 - \tan^{-1} 1) + (\tan^{-1} 3 - \tan^{-1} 2)$$

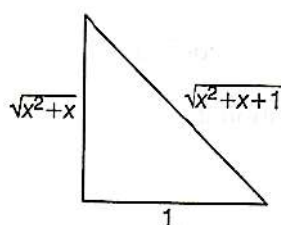
$$+ (\tan^{-1} 4 - \tan^{-1} 3) + \dots + [\tan^{-1}(n+1) - \tan^{-1} n]$$

$$\theta = \tan^{-1}(n+1) - \tan^{-1}(1)$$

$$\theta = \tan^{-1}\left(\frac{n+1-1}{1+n+1}\right) \Rightarrow \tan \theta = \frac{n}{n+2}$$

2. (c) $\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2+x+1} = \frac{\pi}{2}$

$$\cos^{-1} \frac{1}{\sqrt{x^2+x+1}} + \sin^{-1} \sqrt{x^2+x+1} = \frac{\pi}{2}$$



$$\cos^{-1} \frac{1}{\sqrt{x^2+x+1}} = \frac{\pi}{2} - \sin^{-1} \sqrt{x^2+x+1}$$

$$\cos^{-1} \frac{1}{\sqrt{x^2+x+1}} = \cos^{-1} \sqrt{x^2+x+1}$$

$$\frac{1}{\sqrt{x^2+x+1}} = \sqrt{x^2+x+1}$$

$$x^2+x+1=1 \Rightarrow x^2+x=0 \Rightarrow x(x+1)=0$$

$$x=0, x=-1$$

3. (a) $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$

$$\tan^{-1}\left(\frac{5x}{1-6x^2}\right) = \frac{\pi}{4} \Rightarrow \frac{5x}{1-6x^2} = \tan \frac{\pi}{4}$$

$$\Rightarrow 5x = 1 - 6x^2 \Rightarrow 6x^2 + 5x - 1 = 0$$

$$\Rightarrow (x+1)(6x-1) = 0 \Rightarrow x = -1, x = \frac{1}{6}$$

4. (d) $\sin^{-1} x + \cos^{-1}(1-x) = \sin^{-1}(-x)$

$$\Rightarrow 2\sin^{-1} x + \cos^{-1}(1-x) = 0 \left[\because \sin^{-1}(-x) = -\sin^{-1} x \right]$$

$$2\left(\frac{\pi}{2} - \cos^{-1} x\right) + \cos^{-1}(1-x) = 0 \left[\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right]$$

$$\Rightarrow \pi - 2\cos^{-1} x + \cos^{-1}(1-x) = 0$$

$$\Rightarrow \cos^{-1}(1-x) = 2\cos^{-1} x - \pi$$

Let,

$$\cos^{-1} x = \theta$$

$$x = \cos \theta$$

So,

$$\cos^{-1}(1-x) = 2\theta - \pi$$

$$1-x = \cos(2\theta - \pi) \Rightarrow 1-x = \cos(\pi - 2\theta)$$

$$\Rightarrow 1-x = -\cos 2\theta \Rightarrow 1-x = -(2\cos^2 \theta - 1)$$

$$1-x = 1 - 2\cos^2 \theta \Rightarrow x = 2\cos^2 \theta$$

$$\Rightarrow x = 2x^2 \Rightarrow 2x^2 - x$$

$$= 0 \Rightarrow x(2x-1) = 0 \Rightarrow x = 0, x = \frac{1}{2}$$

5. (d) $\sin^{-1}\left(\frac{2a}{1+a^2}\right) - \cos^{-1}\left(\frac{1-b^2}{1+b^2}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$$\therefore \left[\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1} x, \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2\tan^{-1} x \right]$$

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) = 2\tan^{-1} x$$

$$2\tan^{-1} a - 2\tan^{-1} b = 2\tan^{-1} x$$

$$\Rightarrow \tan^{-1}\left(\frac{a-b}{1+ab}\right) = \tan^{-1} x \Rightarrow x = \frac{a-b}{1+ab}$$

6. (b)

$$\therefore \left[\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1} x, \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2\tan^{-1} x \right]$$

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) = 2\tan^{-1} x$$

$$2\tan^{-1} a - 2\tan^{-1} b = 2\tan^{-1} x$$

$$\Rightarrow \tan^{-1}\left(\frac{a-b}{1+ab}\right) = \tan^{-1} x \Rightarrow x = \frac{a-b}{1+ab}$$

$$\cot^{-1}(21) + \cot^{-1}(13) + \cot^{-1}(-8) = ?$$

$$= \tan^{-1}\left(\frac{1}{21}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \tan^{-1}\left(-\frac{1}{8}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{21} + \frac{1}{13}}{1 - \frac{1}{21 \cdot 13}}\right) + \cot^{-1}(-8)$$

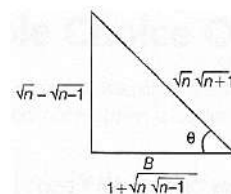
$$= \tan^{-1} \frac{1}{8} + \pi - \cot^{-1} 8 \left[\because \cot^{-1}(-x) = \pi - \cot^{-1} x \right]$$

$$= \cot^{-1} 8 + \pi - \cot^{-1} 8 = \pi$$

7. (c)

$$\sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2}-\sqrt{1}}{\sqrt{6}} + \sin^{-1} \frac{\sqrt{3}-\sqrt{2}}{\sqrt{12}} + \dots$$

$$T_n = \sin^{-1} \frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n}\sqrt{n+1}} = \theta \Rightarrow \frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n}\sqrt{n+1}} = \sin \theta$$



$$\text{Base}^2 = n(n+1) - (\sqrt{n} - \sqrt{n-1})^2$$

$$= n(n+1) - (n + n - 1 - 2\sqrt{n(n-1)})$$

$$= n^2 + n - 2n + 1 + 2\sqrt{n(n-1)}$$

$$= n^2 - n + 1 + 2\sqrt{n(n-1)}$$

$$\text{Base} = \sqrt{n^2 - n + 1 + 2\sqrt{n}\sqrt{n-1}}$$

$$= \sqrt{n(n-1) + 1 + 2\sqrt{n}\sqrt{n-1}}$$

$$= \sqrt{\sqrt{n(n-1)}^2 + (1)^2 + 2\sqrt{n}\sqrt{n-1}} = \sqrt{n(n-1)} + 1$$

$$\text{So, } \tan \theta = \frac{\sqrt{n}-\sqrt{n-1}}{1+\sqrt{n}\sqrt{n-1}} \Rightarrow \theta = \tan^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{1+\sqrt{n}\sqrt{n-1}}\right)$$

$$\theta = \tan^{-1} \sqrt{n} - \tan^{-1} \sqrt{n-1} = T_n$$



$$T_1 = \tan^{-1} \sqrt{1} - \tan^{-1} 0 \Rightarrow T_2 = \tan^{-1} \sqrt{2} - \tan^{-1} \sqrt{1}$$

$$T_3 = \tan^{-1} \sqrt{3} - \tan^{-1} \sqrt{2}$$

$$T_\infty = \tan^{-1} \sqrt{n} - \tan^{-1} \sqrt{n-1}$$

$$\text{Sum} = \tan^{-1} \sqrt{n} - \tan^{-1} 0 \Rightarrow n \rightarrow \infty$$

$$\text{Sum} = \tan^{-1} \sqrt{\infty} = \tan^{-1} \infty = \frac{\pi}{2}$$

8. (d) $\cot^{-1}(-x) = \pi - \cot^{-1} x$

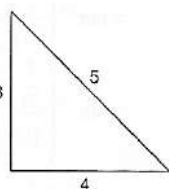
$$= \cot^{-1}(-\sqrt{3}) = \pi - \cot^{-1} \sqrt{3}$$

$$= \pi - \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

9. (a) $\cot \left(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3} \right) = \cot \left(\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3} \right)$

$$= \cot \left[\tan^{-1} \left(\frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{2}{12}} \right) \right]$$

$$= \cot \left(\tan^{-1} \frac{17}{6} \right)$$



$$= \cot \left(\tan^{-1} \frac{17}{6} \right) = \cot \left(\cot^{-1} \frac{6}{17} \right) = \frac{6}{17}$$

10. (c) $2 \tan^{-1} [\operatorname{cosec}(\tan^{-1} x) - \tan(\cot^{-1} x)]$

$$= 2 \tan^{-1} \left[\operatorname{cosec} \left(\operatorname{cosec}^{-1} \frac{\sqrt{1+x^2}}{x} \right) \right] - \tan \left(\tan^{-1} \frac{1}{x} \right)$$

$$= 2 \tan^{-1} \left[\frac{\sqrt{1+x^2}-1}{x} \right]$$

Put $x = \tan \theta$

$$= 2 \tan^{-1} \left[\frac{\sec \theta - 1}{\tan \theta} \right] = 2 \tan^{-1} \left[\frac{1 - \cos \theta}{\sin \theta} \right]$$

$$= 2 \tan^{-1} \left[\tan \frac{\theta}{2} \right] = 2 \times \frac{\theta}{2} \Rightarrow \theta = \tan^{-1} x$$

11 (a) Refer to solution 9.

12. (b) $\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} = \phi$

$$\cos \left(\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} \right) = \cos \phi$$

$$\cos \left[\cos^{-1} \left(\frac{x}{2} \right) \right] \cos \left[\cos^{-1} \left(\frac{y}{3} \right) \right]$$

$$- \sin \left[\cos^{-1} \frac{x}{2} \right] \sin \left[\cos^{-1} \frac{y}{3} \right] = \cos \phi$$

$$\frac{x}{2} \cdot \frac{y}{3} - \sin \left(\sin^{-1} \sqrt{1 - \frac{x^2}{4}} \right) \sin \left(\sin^{-1} \sqrt{1 - \frac{y^2}{9}} \right) = \cos \phi$$

$$\frac{xy}{6} - \sqrt{1 - \frac{x^2}{4}} \sqrt{1 - \frac{y^2}{9}} = \cos \phi$$

$$\frac{xy}{6} - \cos \phi = \sqrt{1 - \frac{x^2}{4}} \sqrt{1 - \frac{y^2}{9}}$$

$$\frac{x^2 y^2}{36} + \cos^2 \phi - \frac{xy \cos \phi}{3} = \left(1 - \frac{x^2}{4} - \frac{y^2}{9} + \frac{x^2 y^2}{36} \right)$$

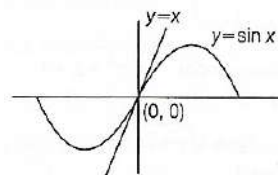
$$9x^2 + 4y^2 - 12xy \cos \phi = 36 \sin^2 \phi$$

13 (b) $\cos^{-1}(-x) = \pi - \cos^{-1} x$

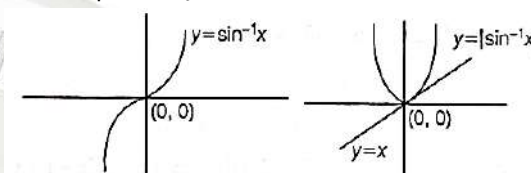
14. (c) Refer to solution 2.

15 (b) $x = \sin x$

We should know that $\sin x < x < \tan x$ when $x \in \left(0, \frac{\pi}{2} \right)$



Hence, $n_2 = 1$ (It has only one solution that (0,0)) We should know that graph of $f^{-1}(x)$ is mirror image of $f(x)$ with respect to $y = x$ as a mirror.



Number of solutions $n_1 = 1$ (only (0,0) is the solution of on this equation) of Eq. $x = |\sin^{-1} x|$

Hence, $n_2 - n_1 = 0$.